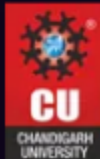


byte{Built 1.0

INNOVATE. IDEATE. INSPIRE.

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Discover. Learn. Empower.

PROBLEM STATEMENT & TEAM DETAILS

TEAM NAME (AS SUBMITTED ON PORTAL)		ArcLight	
PROBLEM STATEMENT TITLE		AI-Powered Process & Document Automation for Business	
S.No	NAME	GENDER	AREA OF EXPERTISE
01	Deven (Team Leader)	Male	Research and Development
02	Rishabh Verma	Male	Model Training
03	Aditya Singh	Male	Backend Developer
04	V Naveen Shankar	Male	Frontend Developer

D-Automation

Proposed Solution:

D-Automation is an AI-native content platform built for the full lifecycle of business and academic documentation. From a single structured prompt, it plans, drafts, and formats presentations, assignments, technical documentation, patent drafts, and research papers – each rendered in the correct professional format, not just as slides. Where tools like Gamma stop at generating a deck, D-Automation extends the same one-prompt-to-polished-document idea across every format a business, student, or researcher actually needs to ship.

OUTLINE OF UNIQUE & INNOVATIVE SOLUTION

What D-Automation Does

- Turns one prompt into a ready, submission-quality output – a slide deck, assignment, report, patent draft, or research paper – not slides alone, like Gamma.

What Makes It Different

- Multi-output engine: one content brief renders into PPTX, DOCX, and PDF, each in the correct professional structure instead of one generic layout.
- Domain-aware templates: built-in structures for research papers, patent claim drafting, academic assignments, and corporate reports – not generic bullet slides.
- Prompt-to-polish pipeline: outline → draft → human review → auto-formatted final file, keeping a person in the loop instead of a blind one-shot generation.
- Institution/brand template matching: upload a college or company template once; every future document inherits its fonts, colors, and layout automatically – the same template-fitting challenge we solved by hand while building this very deck.
- Built-in citation and originality checks for research and patent content, cutting down manual literature-review time.

TECHNICAL APPROACH

Architecture

- Input layer: a structured brief captured via a web dashboard — document type, target template, topic, and depth.
- Orchestration layer: an LLM-based planner breaks the brief into an outline, then generates content section by section.
- Retrieval layer (RAG): a template store and citation index ground research and patent content in verifiable sources.
- Rendering layer: dedicated generators per format — a slide renderer, a document renderer for DOCX/PDF, and a structured renderer for patent and paper formatting.
- Review layer: an in-app chat lets the user refine tone, depth, or structure before final export.

Tech Stack

- Frontend: web dashboard with live document and slide preview.
- Backend: Python/Node microservices orchestrating LLM API calls.
- Document engines: PPTX, DOCX, and structured PDF generation libraries.
- Storage & retrieval: vector database for templates, citations, and style matching.

FEASIBILITY & VIABILITY

Technical Feasibility

- Built on mature, production-ready components — LLM APIs and established document-generation libraries — so an MVP needs no unproven research.
- Modular architecture lets each document type ship independently and improve in parallel.

Market Feasibility

- Businesses, students, and researchers already pay for fragmented single-purpose tools; D-Automation consolidates that spend into one platform.
- Rising enterprise demand for AI-assisted document workflows makes this a timely, addressable market.

Risks & Mitigation

- Risk: factual or legal accuracy in patent and research drafts → Mitigation: retrieval-grounded generation plus mandatory human review before export.
- Risk: formatting breaking on complex templates → Mitigation: a template-validation layer, tested against real institutional templates.

IMPACT & BENEFITS

For Individuals

- Cuts document creation time from hours to minutes for assignments, reports, and presentations.
- Gives students and early researchers access to professional-grade paper and patent drafting support.

For Businesses

- Enforces consistent, on-brand formatting across every team-generated document automatically.
- Frees technical and business teams from repetitive drafting work to focus on higher-value tasks.
- Lowers dependency on costly external design, drafting, and patent-filing support services.

Measurable Outcomes

- Fewer formatting errors and rejected submissions, since every output follows a validated template from the first draft.
- Faster iteration cycles: reviewers see a polished draft immediately, so feedback happens on content, not layout.

BUSINESS MODEL

Revenue Streams

- Freemium SaaS: a free tier with limited monthly documents; Pro and Team subscriptions unlock unlimited generation and advanced templates.
- Premium drafting add-on: a higher-priced tier for patent and research-paper drafting, reflecting the specialized value delivered.
- Enterprise licensing: custom-branded template packs and private deployments for universities and companies.
- API-as-a-service: usage-based pricing for platforms that embed the generation engine.

Indicative Pricing Tiers

- Free — limited documents/month, standard templates, watermark on export.
- Pro (individual) — unlimited generation, premium templates, priority processing.
- Team/Enterprise — custom branding, shared template libraries, admin controls, API access.

Go-To-Market

- Launch with students and campus communities first to build usage and feedback.
- Expand into SME and enterprise accounts through template-licensing partnerships.

RESEARCH & REFERENCES

Prior Art & Inspiration

- Reviewed existing AI presentation generators such as Gamma to identify the gap: strong at slides only, with no support for research papers, patents, or formal assignments. ([WIPO Patent Drafting Manual](#), [\[2005.11401\] Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks](#), [Generative AI: Navigating intellectual property](#))
- Studied the standard formats the platform must reproduce: research-paper structure, patent-claim drafting conventions, and common academic assignment/report rubrics.

Techniques Referenced

- Retrieval-Augmented Generation (RAG) for grounding AI output in verified templates and citations.
- LLM-based structured-document generation for outline-to-draft pipelines.
- Template-matching and style-transfer approaches for reproducing institutional branding.

Repository

- [Dehengoyal885/D-Automation](#)
- **Note:** D-Automation is designed to be significantly different from existing solutions like Gamma. Its key differentiators and innovation will be discussed in greater detail during the final presentation.

Thank You

Team ArcLight — D-Automation

github.com/Devengoyal885/D-Automation